Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

Antennas

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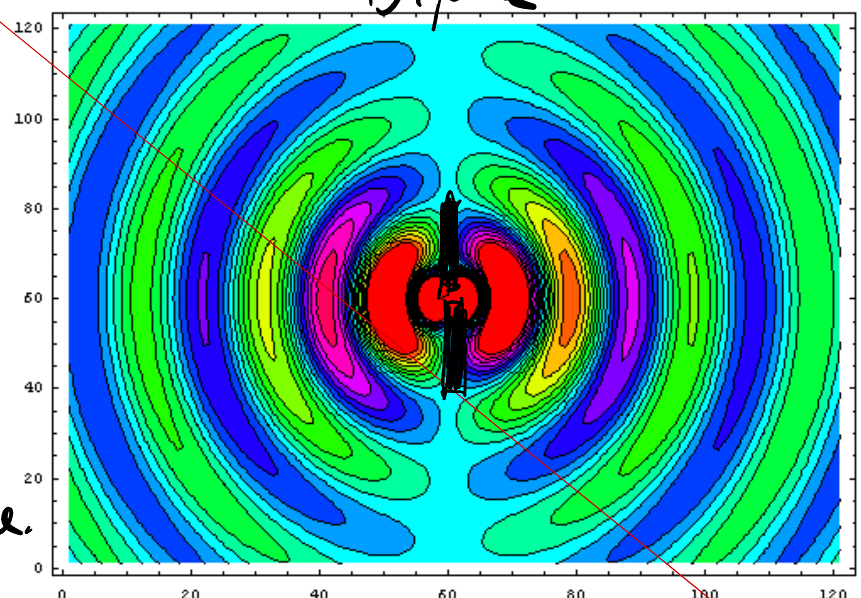
Return loss on the VNA = how much power is bounced back.

- Looked at some basic antenna ideas
  - Matching to free space $\approx 370\Omega$ Impedance of free space
  - Frequencies of operation $\lambda = \frac{v_p}{f}$
  - Returns losses (reflection coefficient) Reflection coefficient : voltage
  - Performances of different types of antenna
    - Dipoles Gain 2.1dBi @ 70Ω
    - Monopoles 5.1dBi
    - Slot
    - Loop
    - Logarithmic
    - Arrays

S_{11}

Power

Dipole



3dB = ×2

double a dipole.

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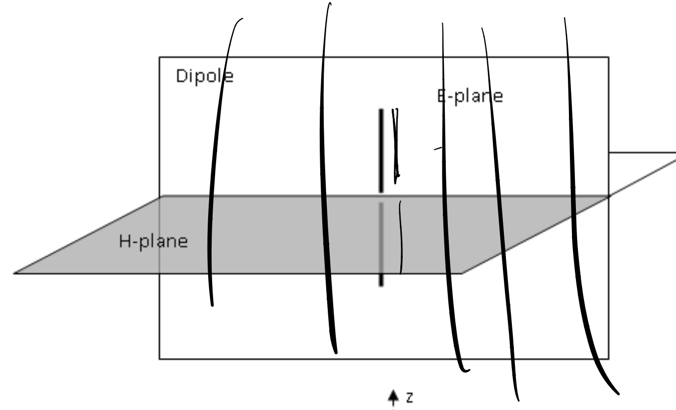
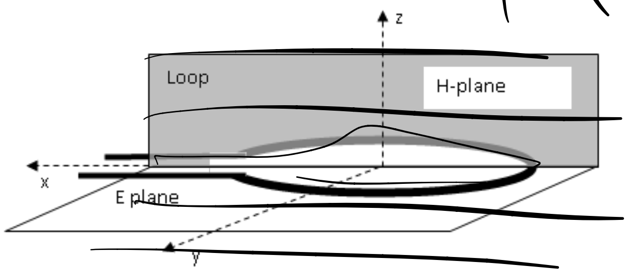
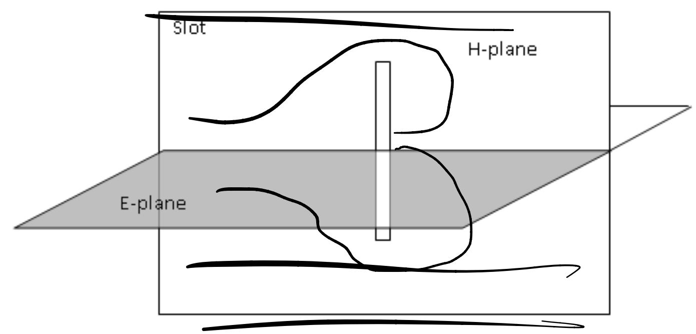
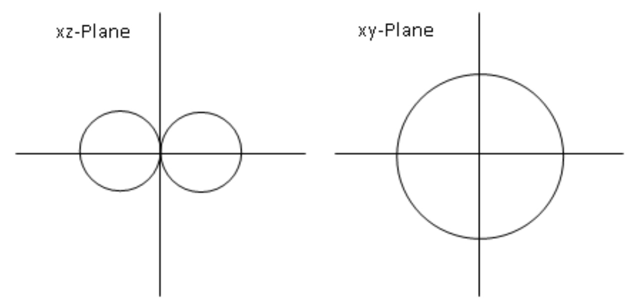
Radiation Patterns

- Dipole
  - Doughnut shaped, 73Ω, 2.1dBi
  - Vertical – vertical polarization
- Loop
  - Similar shape to dipole
  - Pictures shows horizontal – horizontal polarization, vertical – vertical
- Slot
  - Again similar pattern but vertical slot gives horizontal polarization
- Monopole
  - Same pattern but lower half is mirrored by ground hence double the gain









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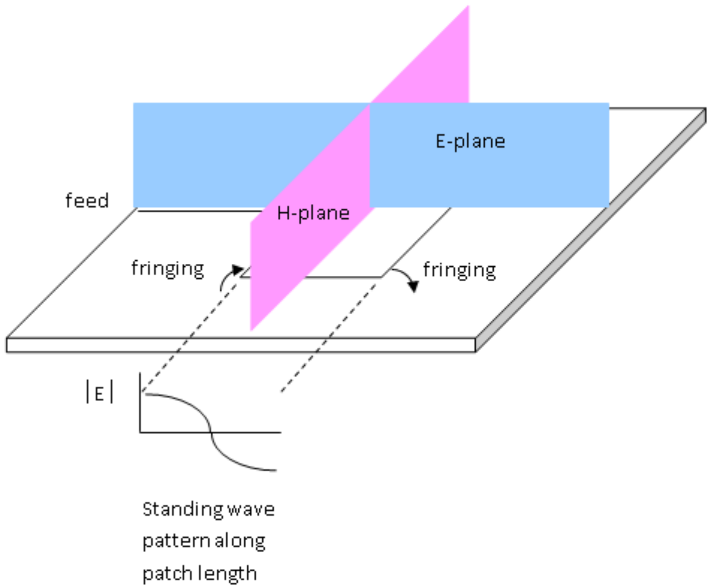
Radiation Patterns

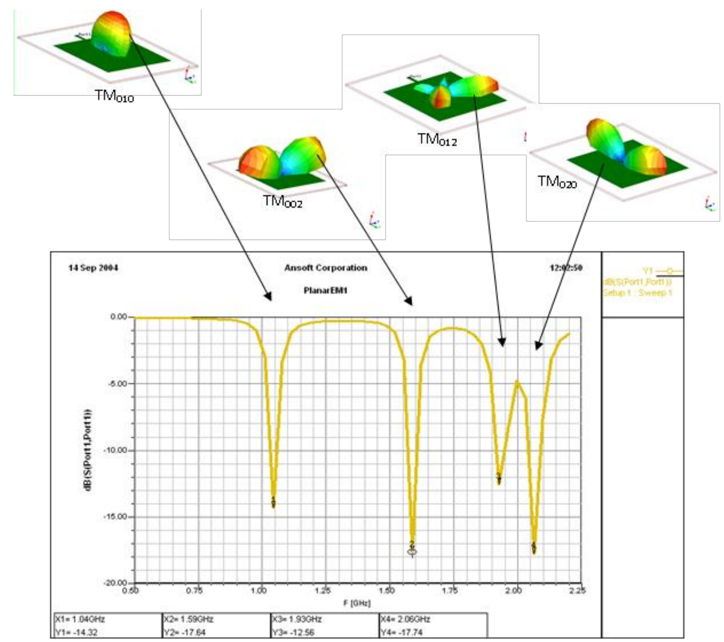


- Patch Antenna
  - Size

$$length = \frac{v}{2f_o} = \frac{3\times10^8}{2f_o\sqrt{\varepsilon_r}}$$

$$width = \frac{\lambda_o}{2}\sqrt{\frac{2}{1+\varepsilon_r}}$$

  - Typically half a wavelength.
    - Linear polarization
  - Radiation pattern



  - http://www.youtube.com/watch?feature=player_detailpage&v=7SuBNGcpr0Q

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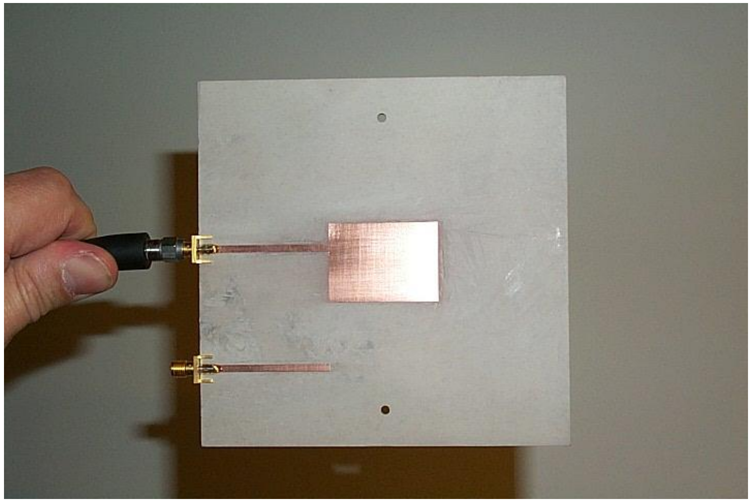
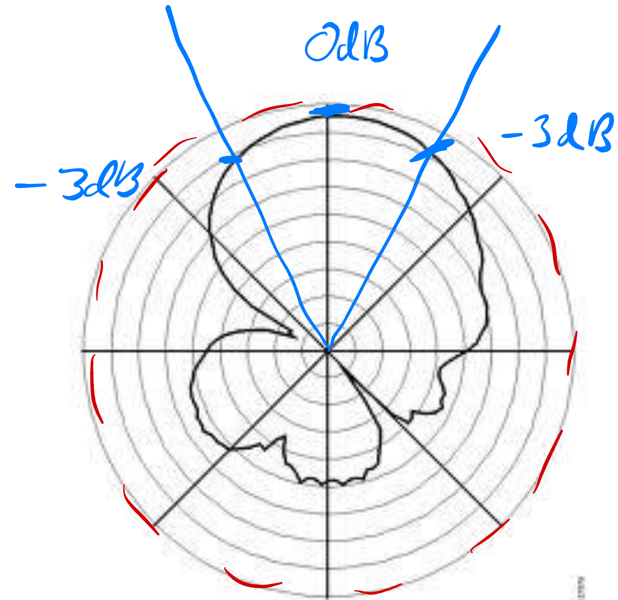
Patch Antenna





Isotropic

Stub match

Antenna pattern is a Sinc function
Side lobe is therefore 13.4dB down

Gain is in dBi

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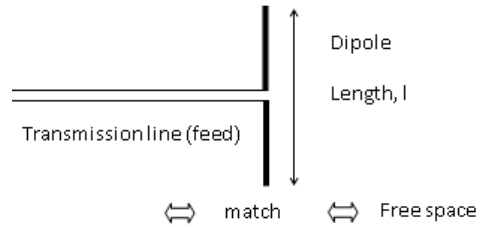
Capacitance and inductance in Antenna

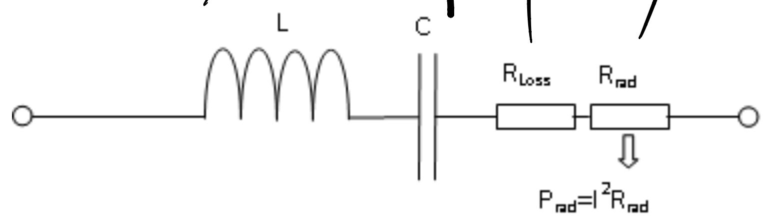
- At low frequencies Antennas are electrically short but physically large.
  - Dipole:



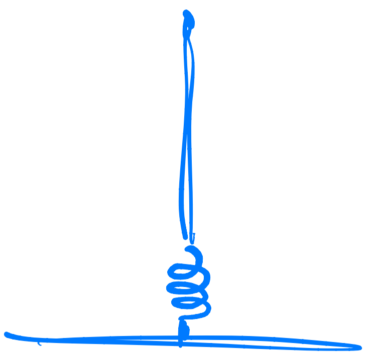
Needs matching

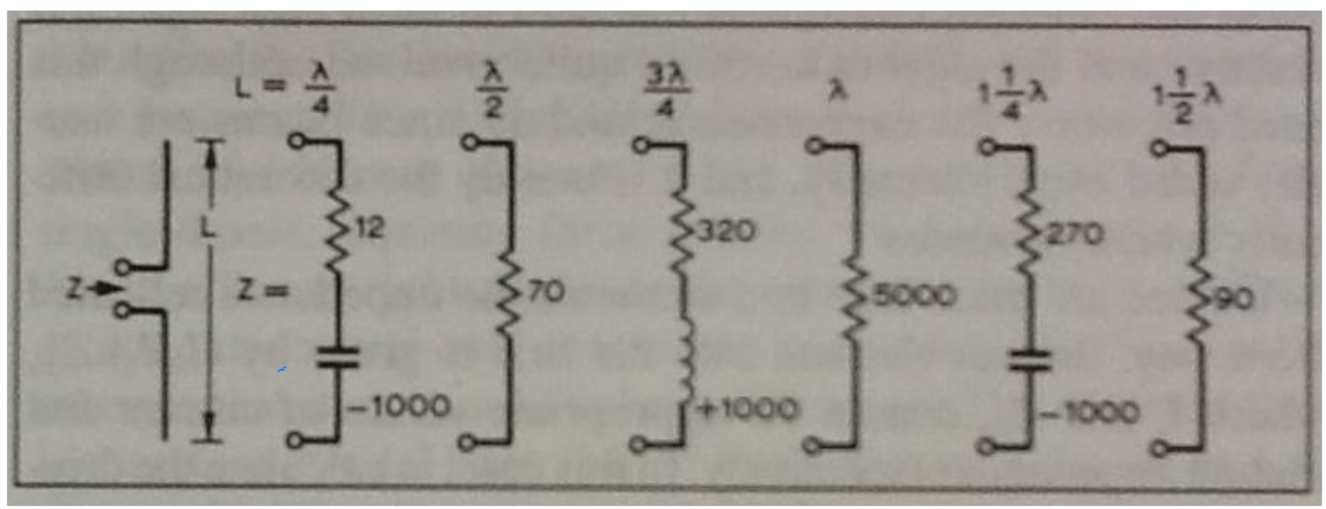
Bad

Good



  - Omnidirectional





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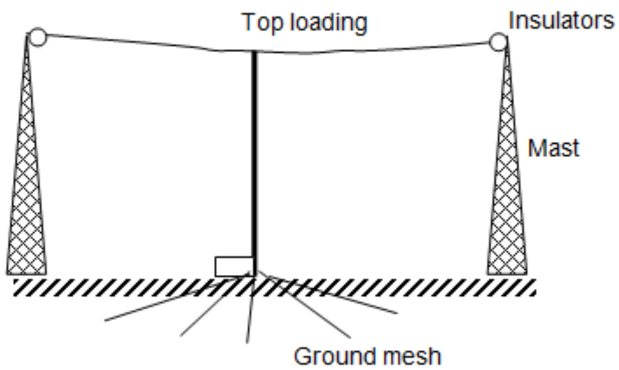
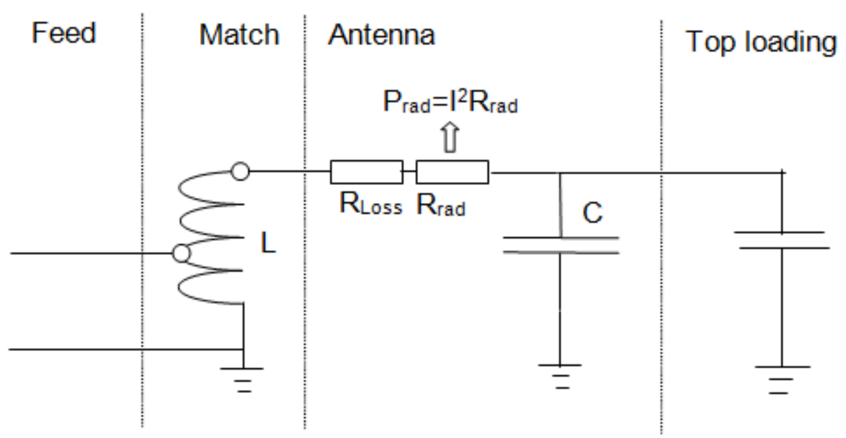
Capacitors and inductors with Antennas





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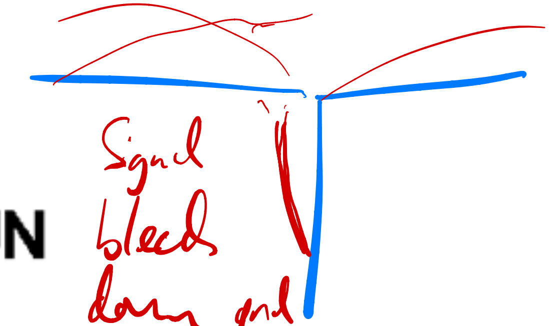
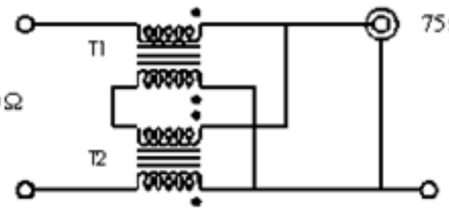
Baluns

- Physical properties of ???:
  - Two (parallel) conductors with either a signal and ground (un-balanced) or the same signal referenced to ground in both, but one is 180 degrees out of phase (balanced).

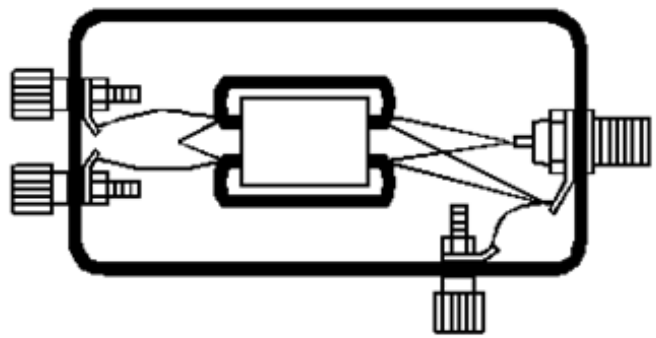
4:1 CHOKE BALUN



Signal bleeds down gnd



- Improves impedance match between antenna and coax.
- Keeps noise on cable shield out of antenna-ground circuit.
- Can be used to couple coax to receivers.
- Covers MF and HF frequency ranges.
- DC path to ground eliminates static build up.



BLN-73-202 FERRITE CORE

FROM AMIDON ASSOCIATES

T1 7 BIFILAR TURNS #26

T2 7 BIFILAR TURNS #26

Antenna Gain

- Effective Aperture: $$A_e = \frac{G\lambda^2}{4\pi}$$
- Gain: $$G = \frac{4\pi A_e}{\lambda^2}$$
- Isotropic antenna has gain = 1 Hence gain increases with frequency.
- Low frequency – low gain due to size.
- High frequency – higher gain.

See paper in lab notes on dle.

- Gain directly from beam width: $$G \approx \frac{4\pi}{\theta_E \theta_H}$$

= radians

Another version for degrees

Antenna performance

- Directivity:

  Directivity is the maximum value of the directive gain given by:

<u>*Radiation intensity from antenna under test*</u> Spot light antenna.

Radiation intensity from isotropic source radiating the same total power Light bulb antenna

- Efficiency in terms of power gain:

Radiated <u>*4π x radiation intensity from antenna*</u>

Power in *Net power accepted by the antenna from a connected source*

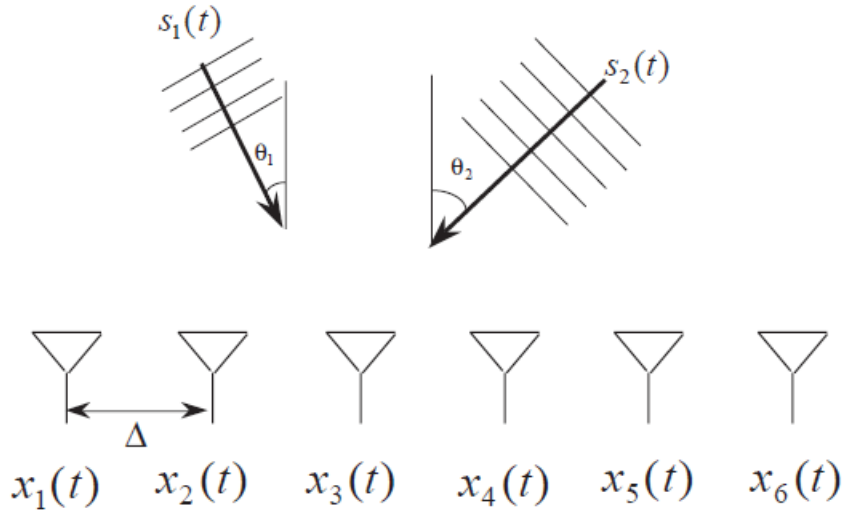
Smart Antennas

- What makes an antenna smart?
  - The ability to gain information about its environment
  - The capability to do something with this information

- All based around phased arrays

An array of simple antenna fed with different phases.

Direction of arrival estimation



Find

1. Number of sources
2. Their direction-of-arrivals (DOAs)
3. Signal Waveforms

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Direction of arrival estimation

Light bulb

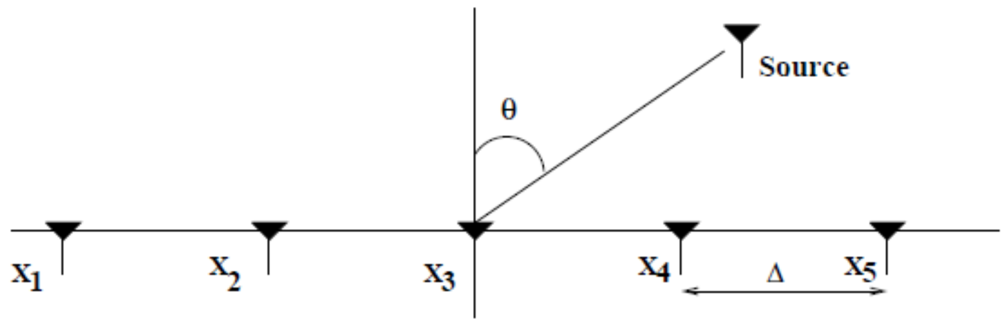
- Assumptions:
  - Isotropic and non-dispersive medium
    - Uniform propagation in all directions
  - Far-Field - Detached from antenna. Radio waves.
    - Radius of propagation >> size of array
    - Plane wave propagation - Straight line
  - Zero mean white noise and signal, uncorrelated
  - No coupling and perfect calibration

Single source, phase and DOA

$$T = \frac{d}{c} \sin\theta$$

Time delay accross the array.

Antenna Array



- Array Response Vector–Far-Field Assumption
  - Delay $\overset{\text{Narrowband}}{\underset{\text{Assumption}}{\Longrightarrow}}$ Phase Shift

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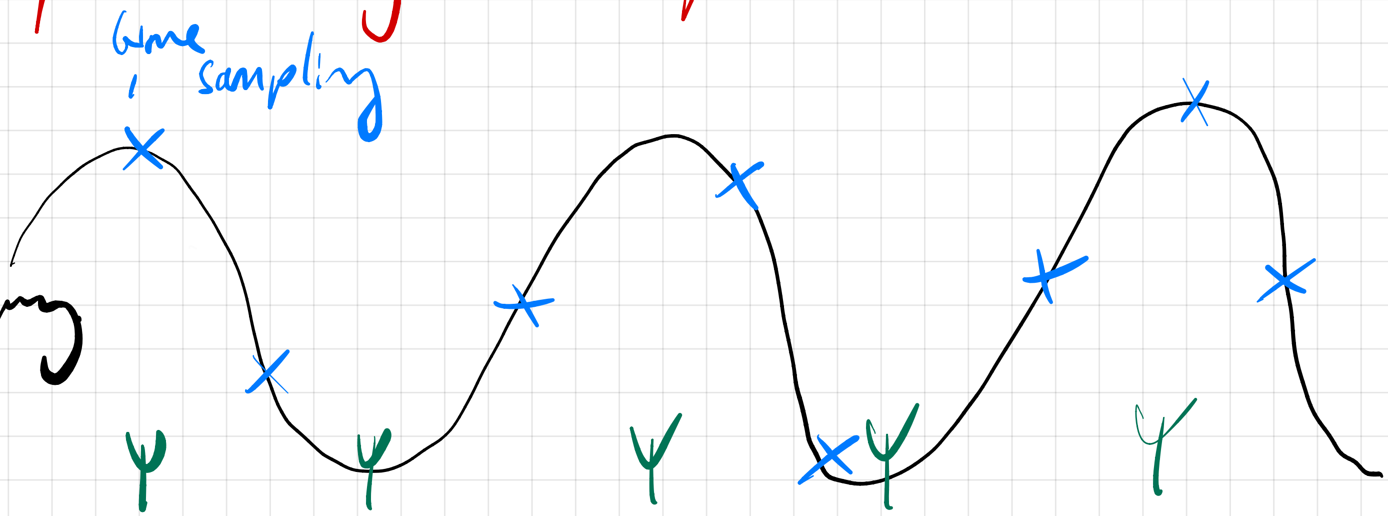
An antenna array samples a signal in space



f_{max} = highest operating frequency

$$2\pi T f_{max} \le \pi$$

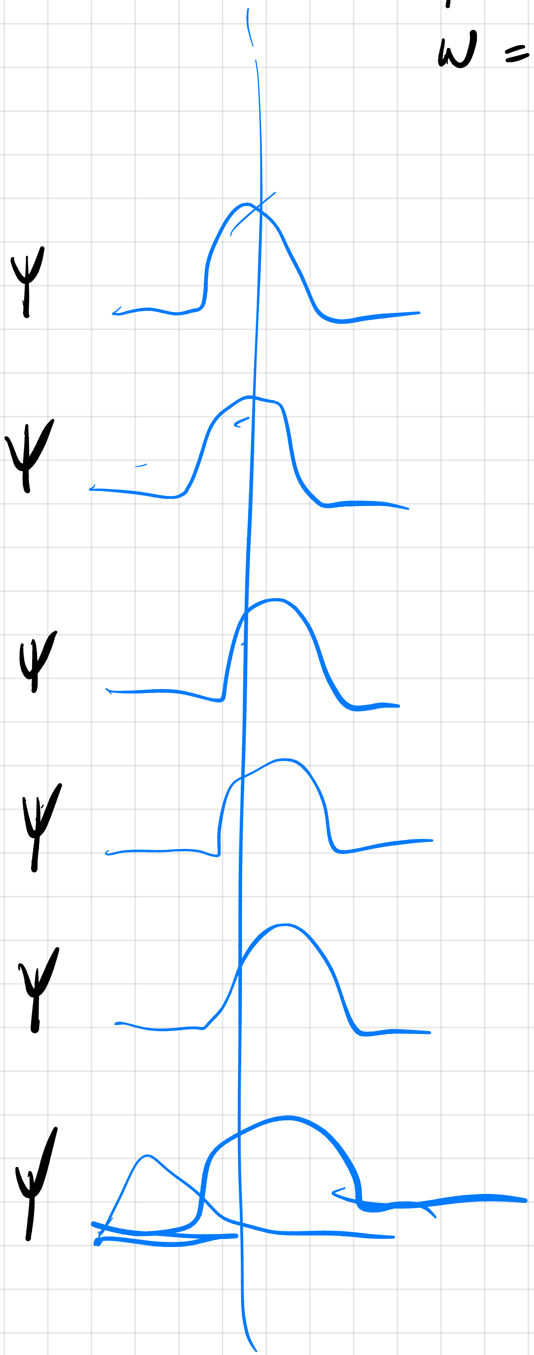
sub $T = d\sin\theta$ $\Rightarrow$ $2\pi \frac{d}{c} \sin\theta f_{max} \le \pi$

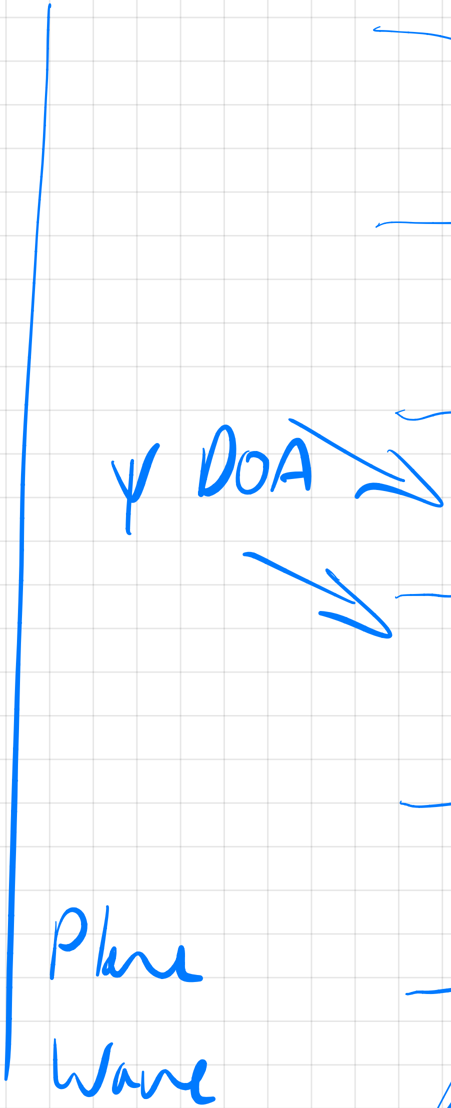
$$2d\sin\theta f_{max} \le c \qquad d \le \frac{c}{2 f_{max} \sin\theta} \qquad d \le \frac{c}{2f}$$

$$d \le \frac{\lambda}{2}$$

Sample in Time

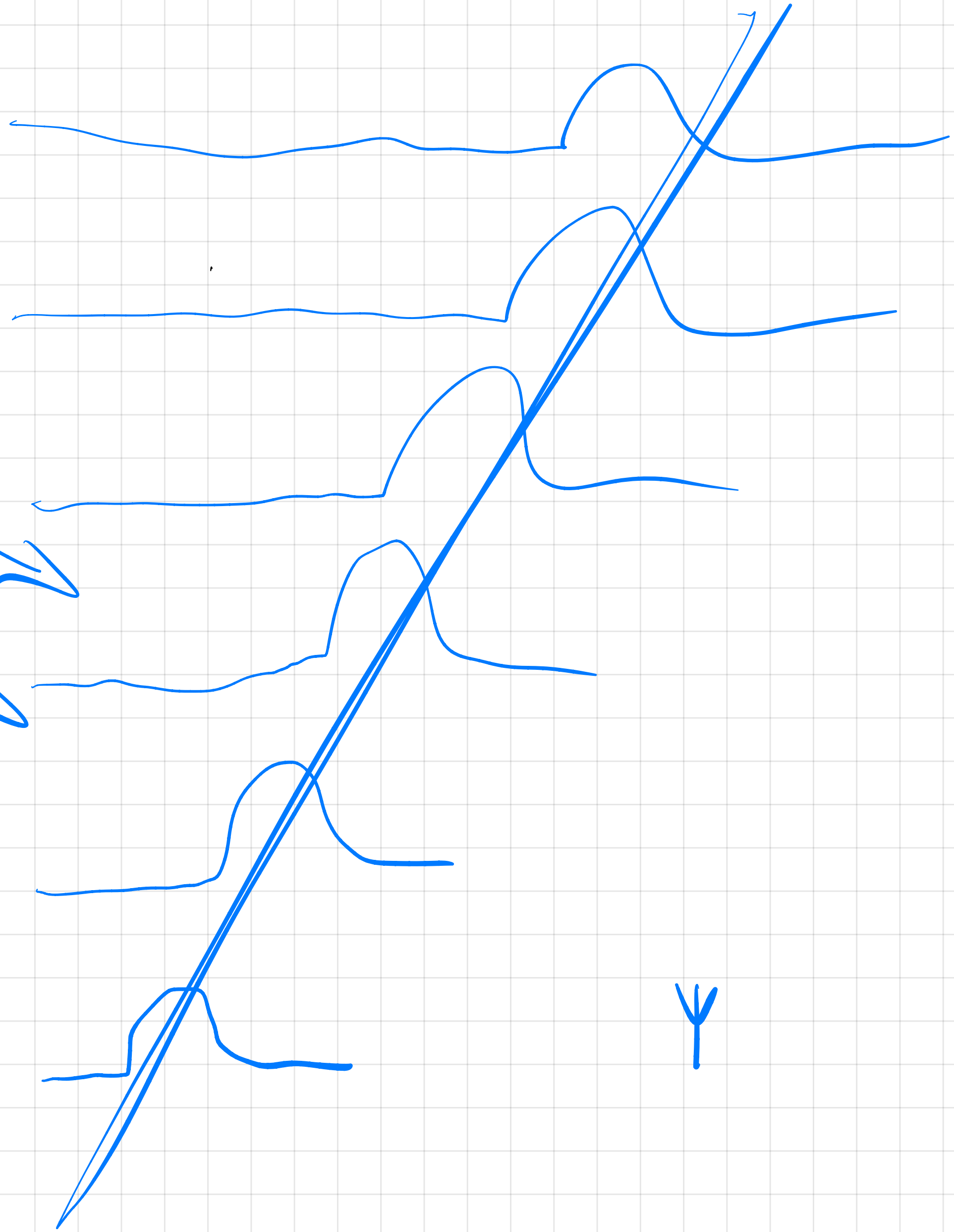
$\omega = \frac{2\pi}{T}$ cycles/sec





In space

$k = \frac{2\pi}{\lambda}$ cycles/metre



Single source, phase and DOA

$$\mathbf{a}(\theta) = [1, e^{j2\pi f_c \triangle \sin\theta/c}, \ldots, e^{j2\pi f_c 4\triangle \sin\theta/c}]^T$$

- Single Source Case $\Longrightarrow \mathbf{x}(t)$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_M(t) \end{bmatrix} = \begin{bmatrix} s_1(t) \\ s_1(t-\tau) \\ \vdots \\ s_1(t-(M-1)\tau) \end{bmatrix} \approx \begin{bmatrix} 1 \\ e^{-j2\pi f_c \tau} \\ \vdots \\ e^{-j2\pi f_c (M-1)\tau} \end{bmatrix} s_1(t) = \mathbf{a}(\theta_1) s_1(t)$$

delay

phase

where $\tau = \triangle \sin\theta_1/c$.

Sampling in space

Multiple source

If you are interested I'll add a paper on this

- By superposition, for d signals,

$$
\begin{aligned}
\mathbf{x}(t) &= \mathbf{a}(\theta_1)s_1(t) + \cdots + \mathbf{a}(\theta_d)s_d(t) \\
&= \sum_{k=1}^{d} \mathbf{a}(\theta_k)s_k(t)
\end{aligned}
$$

Signal sums up & gets better

- Noise

$$
\begin{aligned}
\mathbf{x}(t) &= \sum_{k=1}^{d} \mathbf{a}(\theta_k)s_k(t) + \mathbf{n}(t) \\
&= \mathbf{A}\mathbf{S}(t) + \mathbf{n}(t)
\end{aligned}
$$

Noise averages to zero

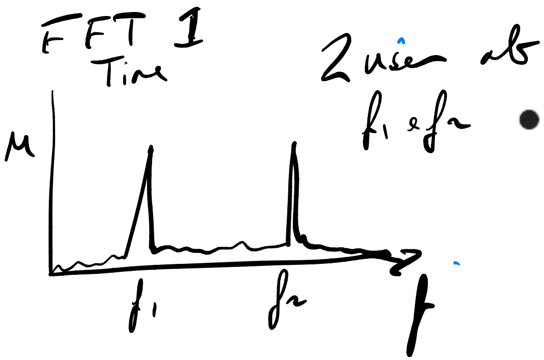
where

$$\mathbf{A} = [\mathbf{a}(\theta_1), \ldots, \mathbf{a}(\theta_d)]$$

and

$$\mathbf{S}(t) = [s_1(t), \ldots, s_d(t)]^T.$$

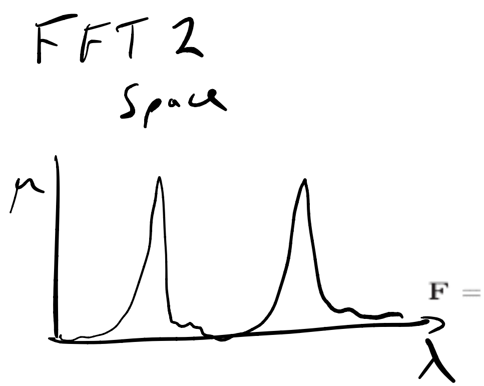
Low resolution approach, beamforming



- Basic Idea

$$x_i(t) = \sum_{k=1}^{d} = e^{(i-1)(j2\pi f_c \triangle \sin\theta_k / c)} s_k(t) = \sum_{k=1}^{d} s_k(t) e^{jw_k(i-1)}$$

where $w_k = 2\pi \triangle \sin(\theta_k)/c$ and $i = 1, \ldots, M$.

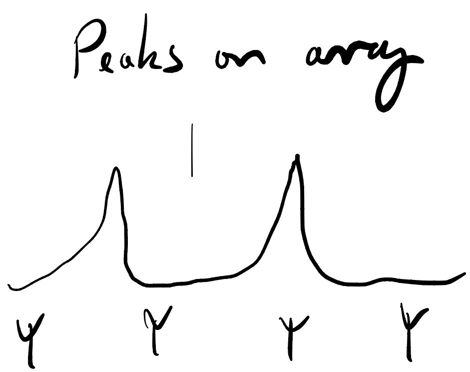


- Use DFT (or FFT) to find the frequencies $\{w_k\}$

Time

$$\mathbf{F} = [\mathbf{F}(w_1) \cdots \mathbf{F}(w_M)] = \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e^{jw_1} & e^{jw_2} & \cdots & e^{jw_M} \\ \vdots & \vdots & \ddots & \vdots \\ e^{j(M-1)w_1} & e^{j(M-1)w_2} & \cdots & e^{j(M-1)w_M} \end{bmatrix}$$

space



- Look for the peaks in

$$|\mathcal{F}(x_i(t))| = |\mathbf{F}^* \mathbf{x}(t)|^2$$

- To smooth out noise

$$B(w_i) = \frac{1}{N} \sum_{t=1}^{N} |\mathbf{F}^* \mathbf{x}(t)|^2$$

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Acting on DOA information

- Beam forming
  - In the receive beamformer the signal from each antenna may be amplified by a different "weight.“
  - Different weighting patterns change the spatial sensitivity of the antenna
  - A main lobe is produced together with nulls
- Main types
  - Zero-forcing (or Null-Steering)
  - Desired signal maximization mode
  - Interference signal minimization or cancellation mode

Beamforming

Switched beamforming

- Fairly simple
    - Several sets of fixed weightings are used.
    - Allows for a choice from a number of beam patterns

Adaptive beamforming

- More complex
    - Weights are changed dynamically in order to steer the main beam onto a target or shift the null onto an interferer
- General improvements:
    - SNR = $\frac{1}{\sigma_n^2} P \cdot L$ Where P= signal power, L = number of elements and σ^2 is noise variance or noise power.

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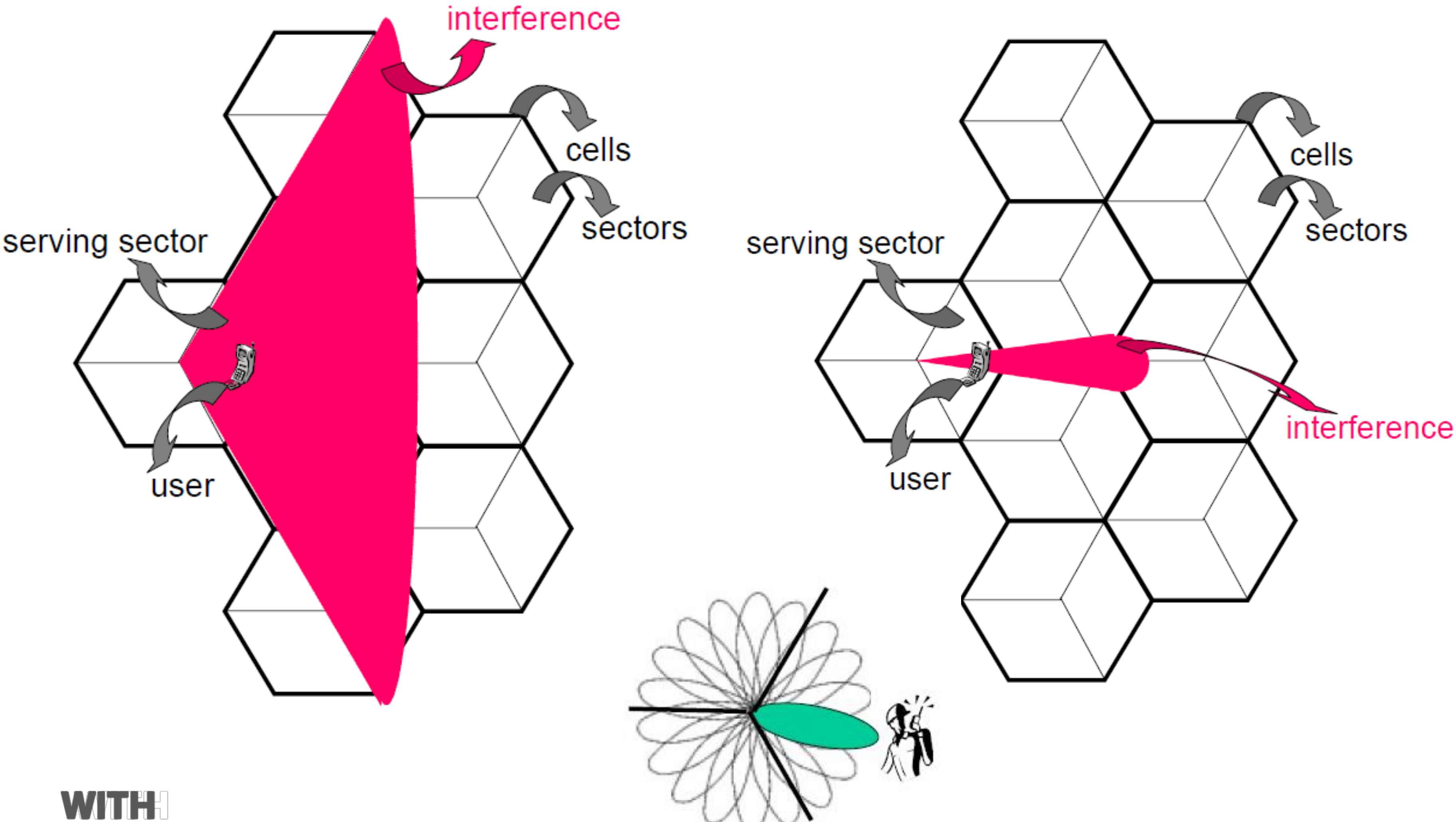
Sectorized and switched beam



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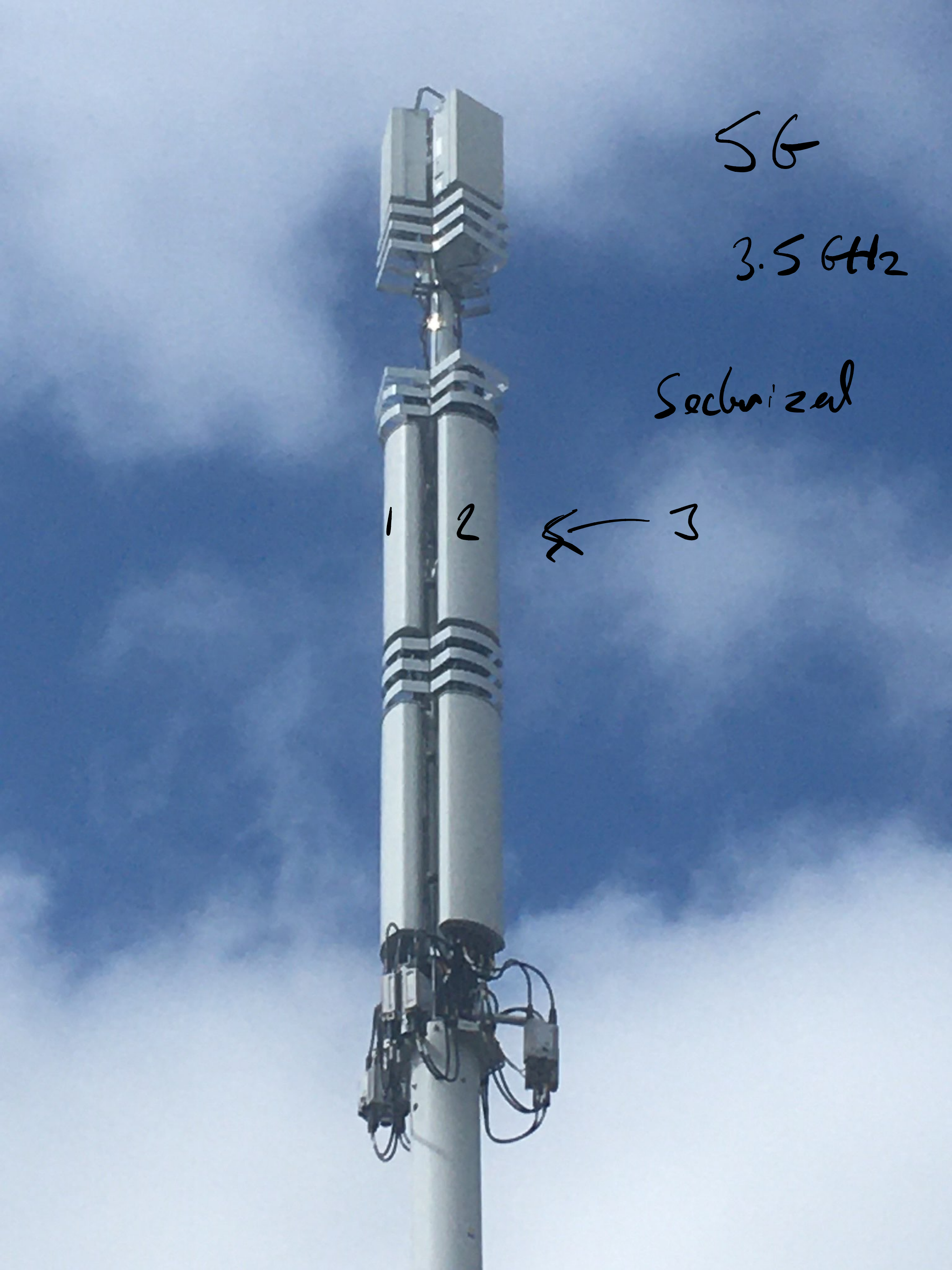
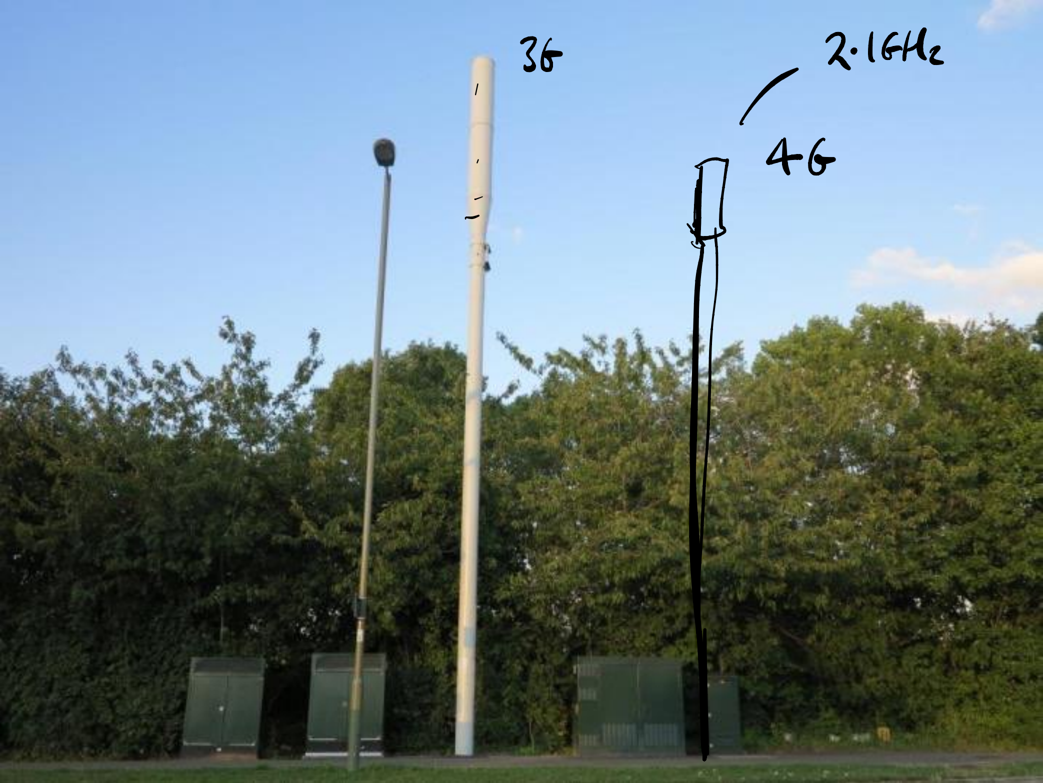

5G
3.5 GHz
Sectorized
1
2
3


5G
goals

Monopole radio
Pt-Pt link
60°
Monopol radio
60°
Sectored

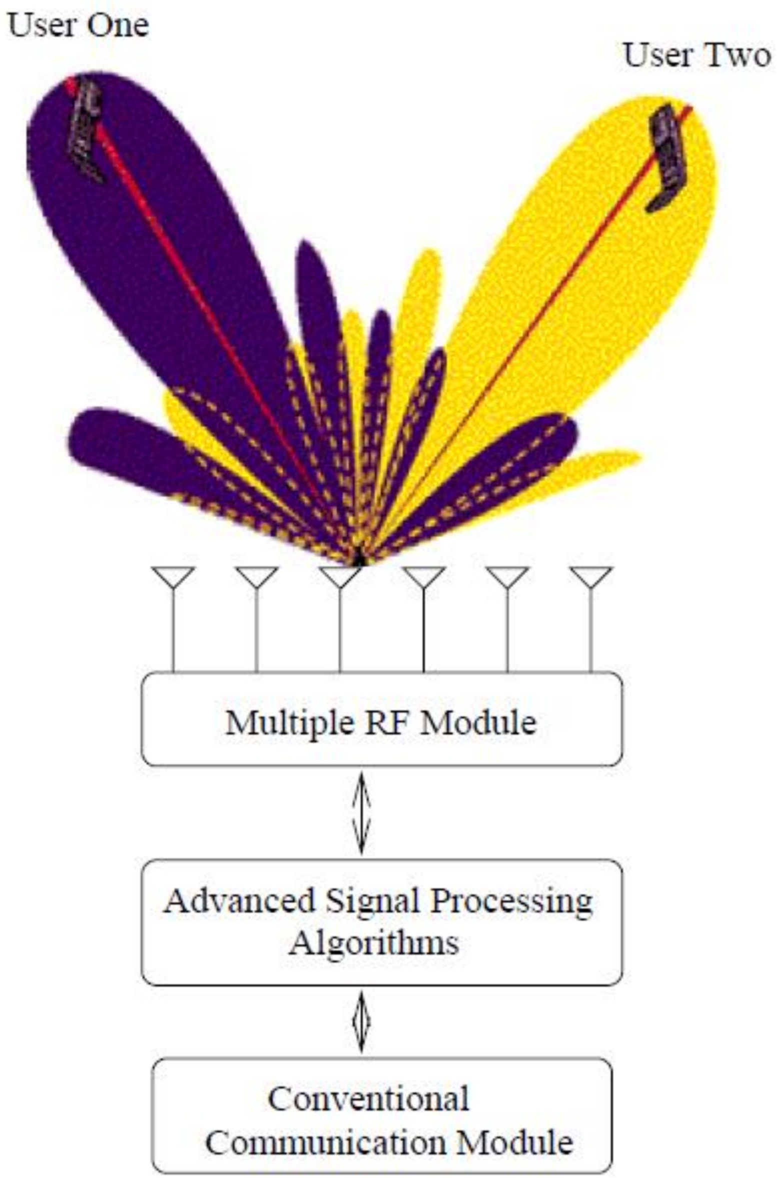

Sectorized.
Each one is an array.


3G
2.1GHz
4G

Macro cell base station

Sectorized.

Example



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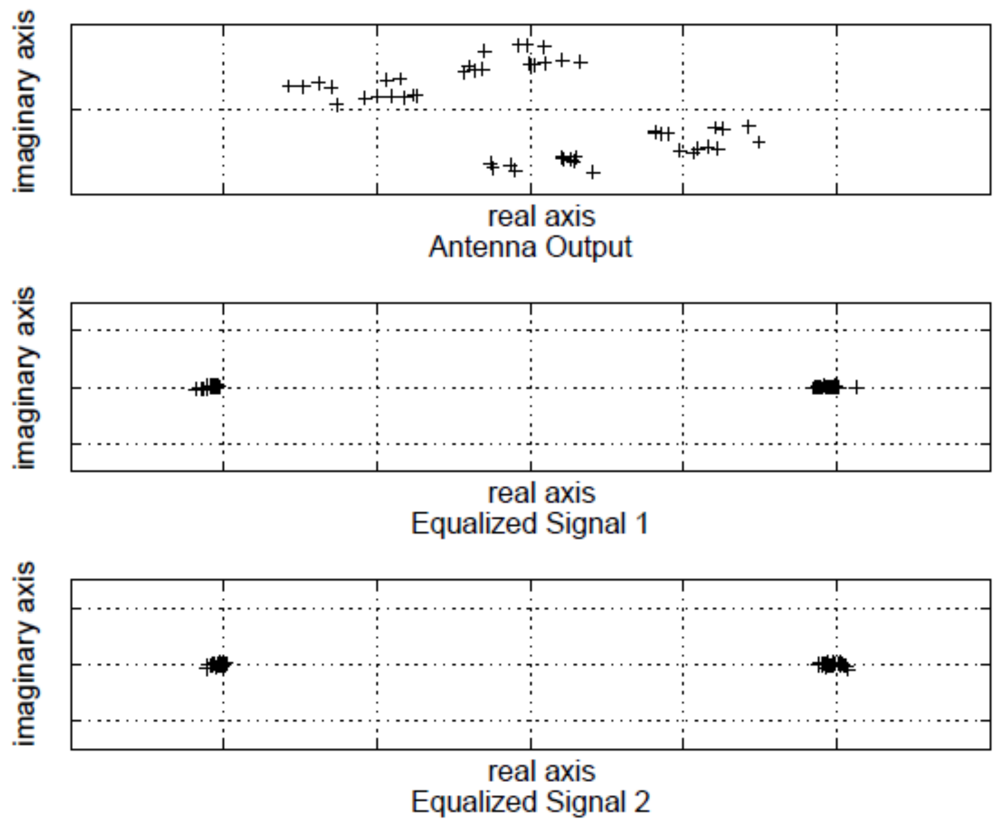
Example

- Comparison of constellation before (upper) and after smart uplink processing (middle and lower)

Phase & weight information (DOA) from array resolves signals

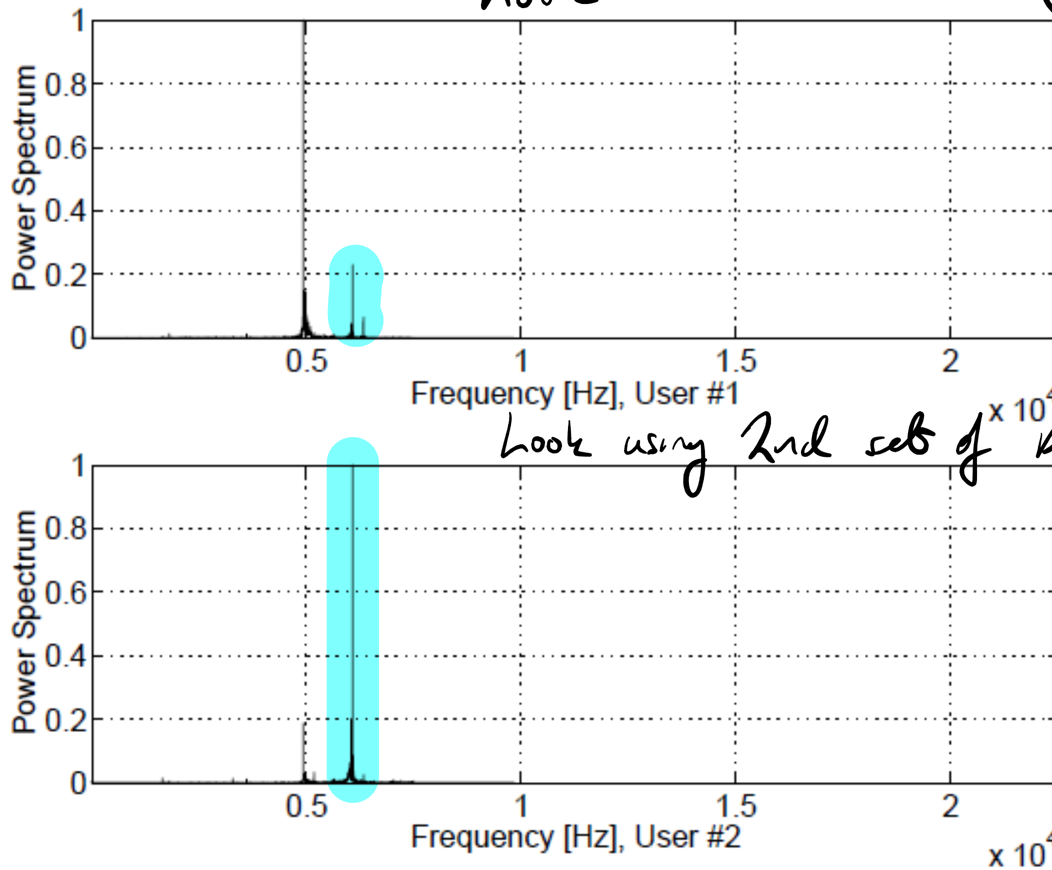


Example

SDMA Spatial division Multiple access

- Beamforming results for two sources separated by 20°

Look at one DOA (set of phase & weights)



Look using 2nd set of DOA information to resolve 2nd user.

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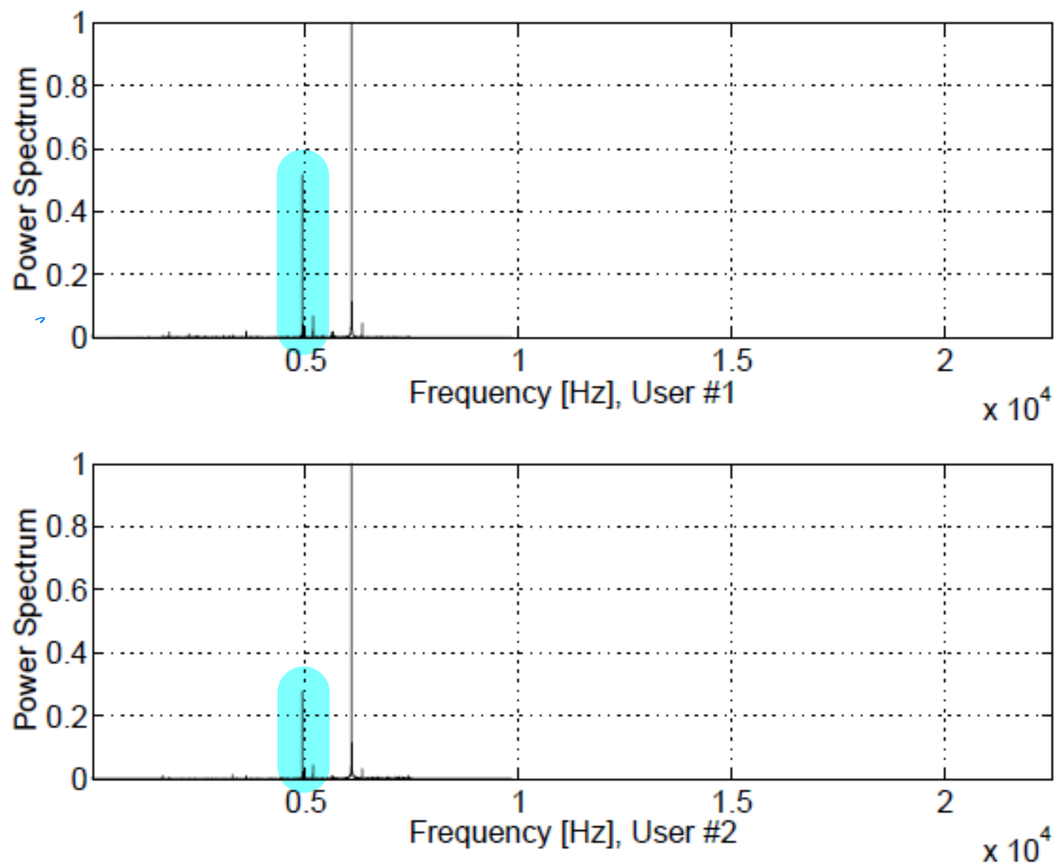
Example

- Beamforming results for two sources separated by 3°

No longer separated enough to beamform



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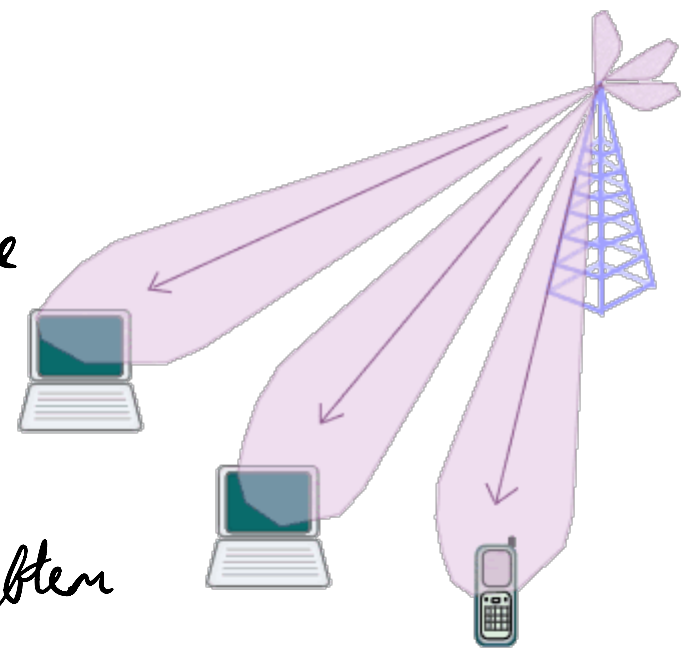
Spatial Division Multiple Access (SDMA)

Spatial division multiple access (SDMA) is a channel access method used in mobile communication systems which reuses the same set of cell phone frequencies in a given service area. Two cells or two small regions can make use of the same set of frequencies if they are separated by an allowable distance (called the reuse distance).



Allows two or more users to be on the same frequency at the same time so long as they are separated in space & the array can resolve them

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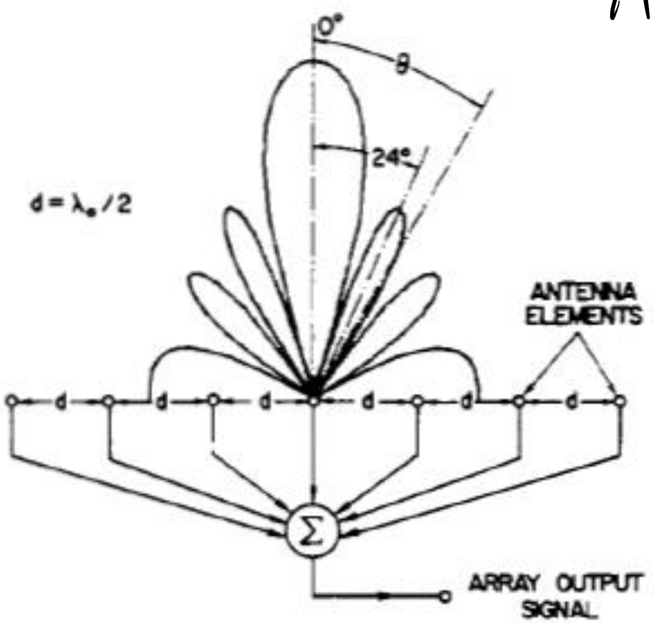
Transmit Beamforming

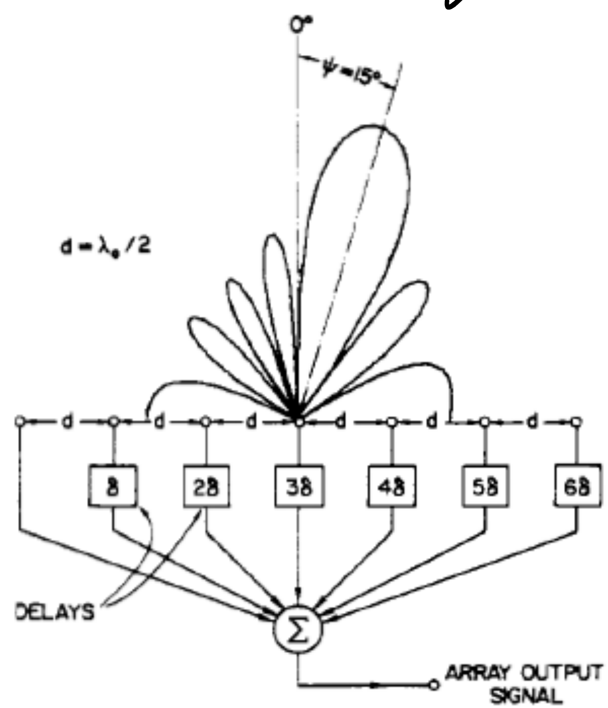
Again either:

- Switched (simple)
- Adaptive (more complex)

Change direction

Adjust phase

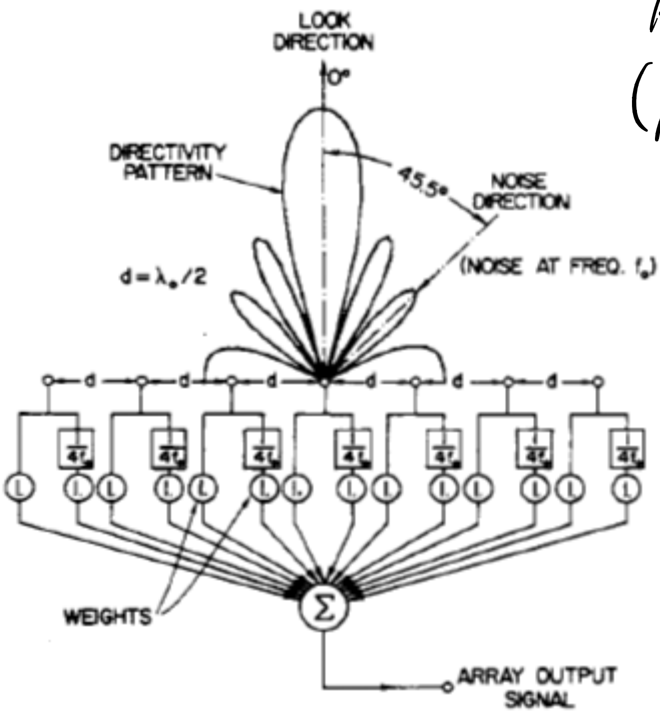


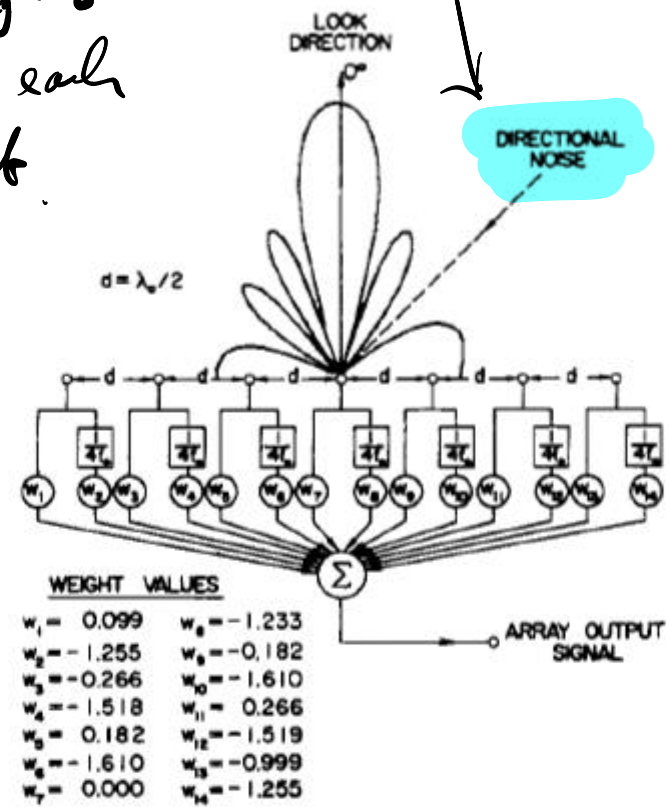


Transmit Beamforming

Change shape.

Adjust weights (power) to each element.





Summary

- Looked at antenna performance
  - Polarization
  - Capacitance
  - Inductance
  - Radiation patterns

Optimum size has just a radiation resistance. Either side will have C or L

- Looked at smart antennas
  - Gaining information, DOA
  - Using it

FFT method

Music - Eigen values & vectors

- Looked at receive beamforming
  - Simple (switched), Clever (adaptive)
- Looked at transmit beamforming
  - The inverse of the receive information used

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